

Experiment 6:

Aim: To demonstrate the performance of knn using wine dataset

Program:

```
# Import libraries
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
import matplotlib.pyplot as plt
import seaborn as sns

# Load Wine dataset
wine = load_wine()

# Features and target
X = wine.data
y = wine.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create KNN model
knn = KNeighborsClassifier(n_neighbors=5)

# Train model
knn.fit(X_train, y_train)

# Predict
y_pred = knn.predict(X_test)

# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:\n", cm)
```

```
# Classification Report
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))
```

```
# Plot Confusion Matrix
plt.figure(figsize=(6,5))
sns.heatmap(cm,
            annot=True,
            cmap="Blues",
            fmt="d",
            xticklabels=wine.target_names,
            yticklabels=wine.target_names)
```

```
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("KNN - Wine Dataset")
plt.show()
```

Accuracy: 0.7407407407407407

Confusion Matrix:

```
[[17  0  2]
 [ 1 15  5]
 [ 1  5  8]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.89	0.89	0.89	19
1	0.75	0.71	0.73	21
2	0.53	0.57	0.55	14
accuracy			0.74	54
macro avg	0.73	0.73	0.73	54
weighted avg	0.74	0.74	0.74	54

